

Problem 11

Problem 1 For the function g defined by

$$g(t) = \frac{t^2 + 7t + 11}{t - 3}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. The function g is a rational function which is continuous on the intervals $(-\infty, 3)$ and $(3, \infty)$. The only candidates for a vertical asymptote is $t = 3$, where g is not defined.

Test of candidate $t = 3$:

$$\lim_{t \rightarrow 3^+} \frac{t^2 + 7t + 11}{t - 3} = \infty$$

Note: the limit is of the form $\frac{\#}{0}$, the numerator positive, and the denominator also positive.

The line $t = 3$ is a vertical asymptote of g since $\lim_{t \rightarrow 3^+} g(t) = \infty$.

- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Explanation. To find the limits, we need to divide the numerator and denominator by the highest power of t in the denominator.

$$\lim_{t \rightarrow \infty} \frac{t^2 + 7t + 11}{t - 3} = \lim_{t \rightarrow \infty} \frac{t + 7 + \frac{11}{t}}{1 - \frac{3}{t}} = \infty$$

Note: the last limit is of the form $\frac{\infty}{\#}$, the numerator positive, the denominator positive.

Therefore, g has no horizontal asymptotes as $t \rightarrow \infty$.

$$\lim_{t \rightarrow -\infty} \frac{t^2 + 7t + 11}{t - 3} = \lim_{t \rightarrow -\infty} \frac{t + 7 + \frac{11}{t}}{1 - \frac{3}{t}} = -\infty$$

Problem 11

Note: the last limit is of the form $\frac{\infty}{\#}$, the numerator negative, the denominator positive.

Therefore, g has no horizontal asymptotes as $t \rightarrow -\infty$ either.